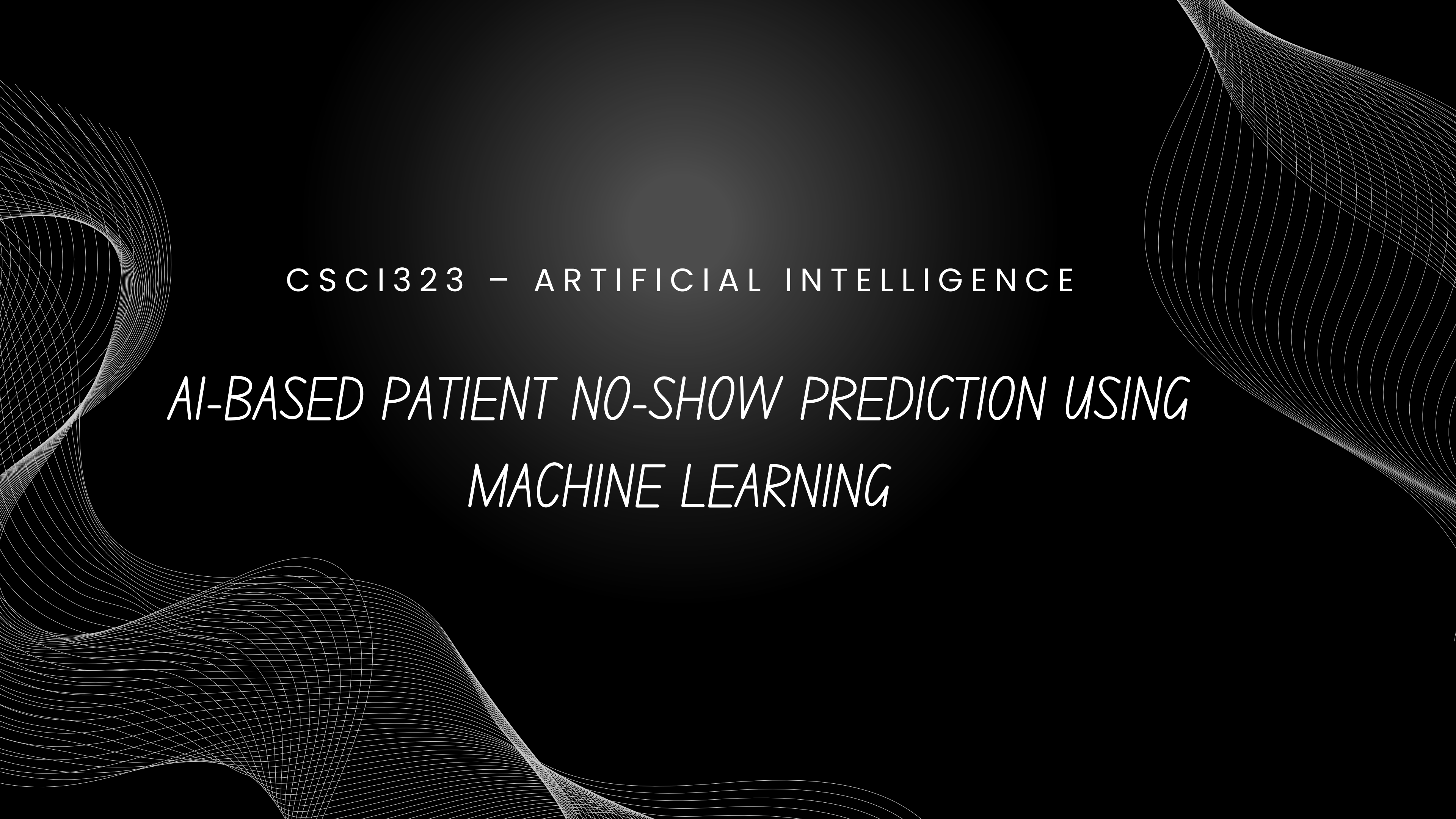




CSCI323 – ARTIFICIAL INTELLIGENCE

*AI-BASED PATIENT NO-SHOW PREDICTION USING
MACHINE LEARNING*

PROBLEM MOTIVATION

The Problem –

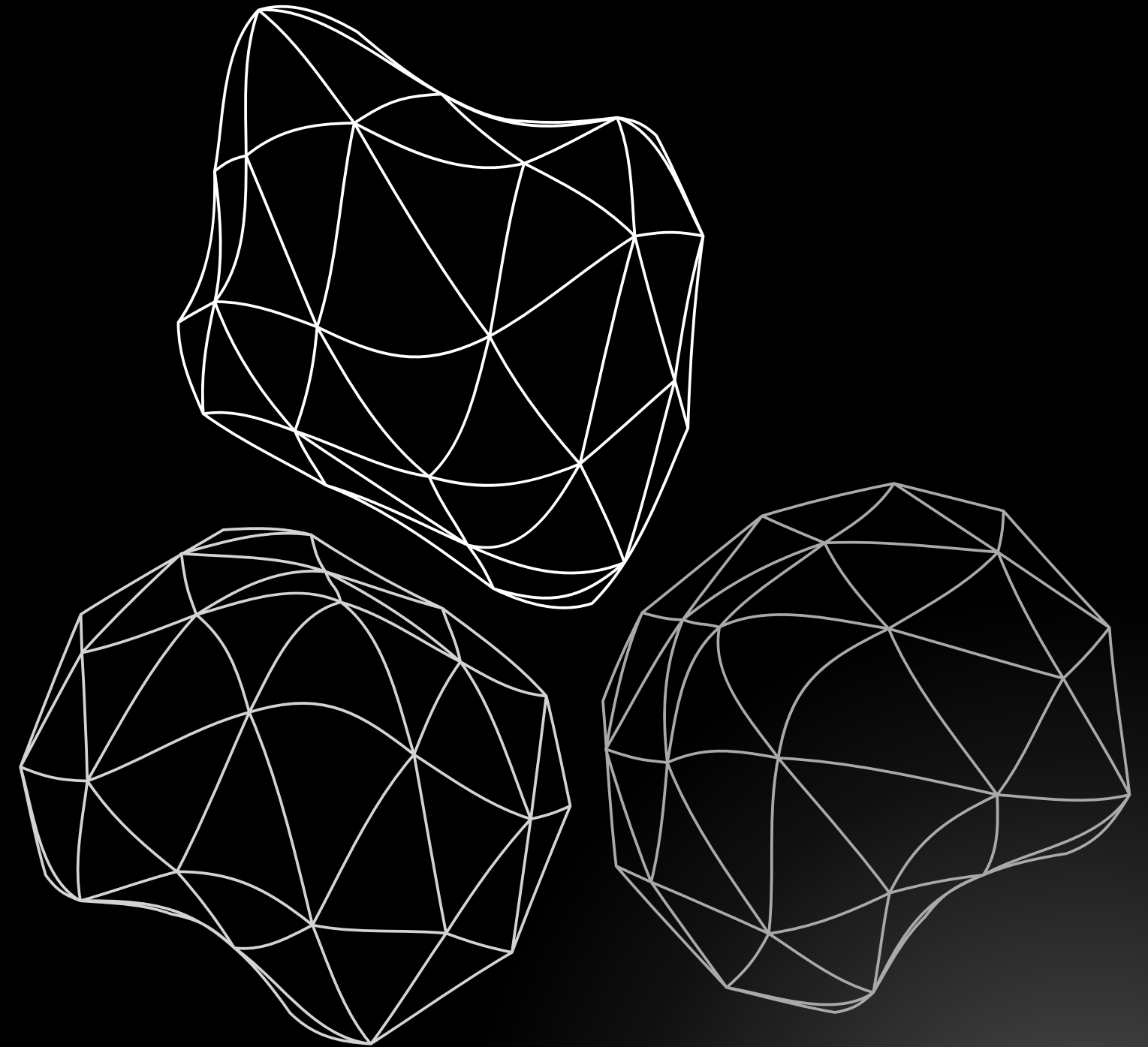
Patient no-shows are a major challenge in hospitals.

When patients miss appointments:

- Appointment slots are wasted
- Doctors' time is unused
- Hospital resources are underutilized
- Waiting times increase for other patients

Dataset –

- 169,524 outpatient appointment records
- 25 medical departments
- UAE healthcare organization



BUSINESS CONTEXT

Why This Project Matters–

The hospital currently cannot predict which patients will miss appointments.

Impact of No-Shows –

- Reduced operational efficiency
- Higher scheduling costs
- Poor resource planning
- Lower service quality

Project Goal –

Use Artificial Intelligence to predict:

- Show (0)
- No-Show (1)

before the appointment date.



AI MODEL SELECTION: RANDOM FOREST CLASSIFIER

- **Enable Architecture:** Uses a "voting pool" of 100 independent decision trees to predict patient attendance (**0 = Show, 1 = No-Show**).
- **Handles Clinical Complexity:** Excellent at managing high-cardinality categorical variables like the 16 distinct departments under **Admitting Department**.
- **Protects Against Overfitting:** Employs **Bootstrap Aggregation (Bagging)** to ensure the model generalizes well to unseen patient data.
- **No Feature Scaling Required:** Natively processes features with different units, evaluating *Age* (years) and *Lead_Time* (days) without mathematical bias.

TECHNICAL PIPELINE & DATA PARTITIONING

- **Feature Engineering (*Lead_Time*)**: Calculated the exact chronological delta between booking dates and appointment dates:

$$\text{Lead_time} = \text{Appointment Date} - \text{Booked Date/Time}$$

- **Categorical Transformation**: Transformed attributes like *Gender*, *OP Type*, and *Clinic Department* using **One-Hot Encoding (*drop_first=True*)** to prevent multi-collinearity.
- The 80/20 Data Split: Divided the 169,524 raw records into:
 - 1. 80% Training Set (135,619 rows): Used to build the model's logic trees.
 - 2. 20% Testing Set (33,905 rows): Kept completely isolated to validate final accuracy.
- **Reproducibility Control**: Locked data rows using a fixed seed (`random_state=42`) to guarantee fair model comparisons.

IMPLEMENTATION

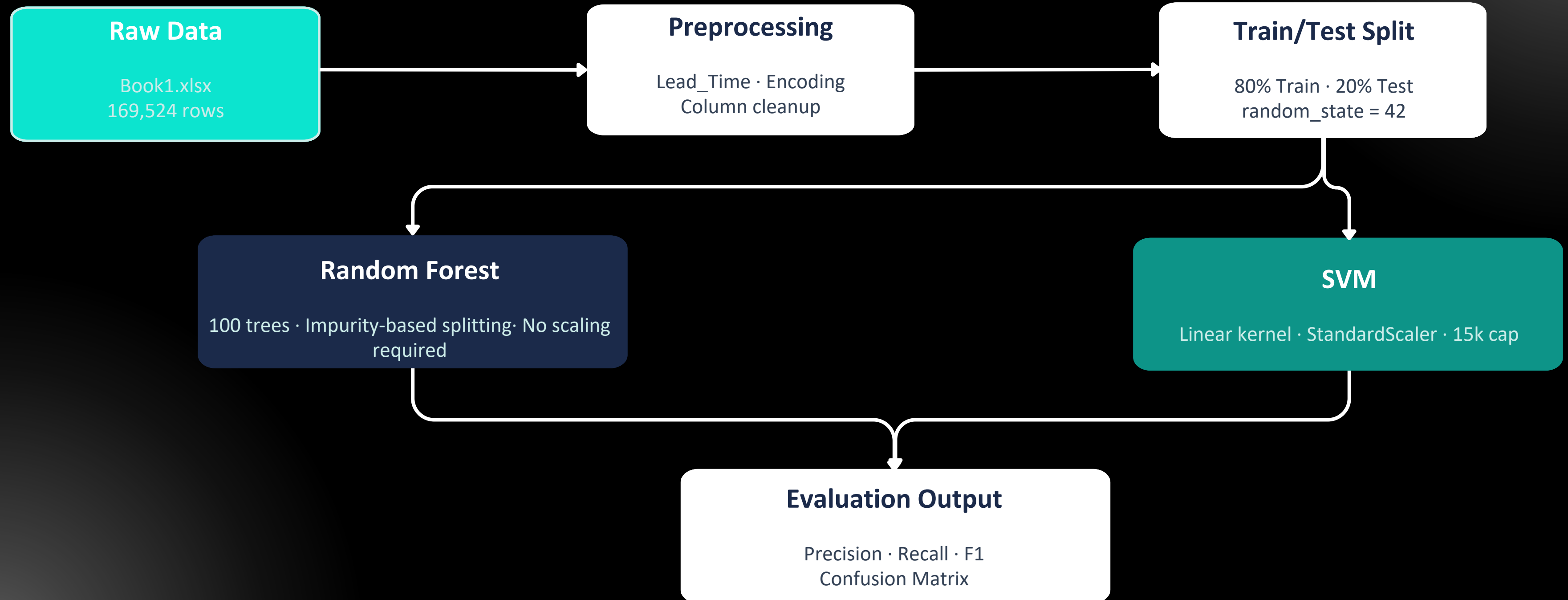
Random Forest

- 1 Feature Engineering**
Lead_Time = Appointment Date - Booking Date (min 0)
- 2 One-Hot Encoding**
Gender, OP Type, New/Revisit, Dept → binary flags
- 3 Train / Test Split**
80% train · 20% test · random_state = 42
- 4 No Scaling Needed**
Tree splits are magnitude-agnostic
- 5 Model Training**
RandomForestClassifier(n_estimators=100, random_state=42)

Support Vector Machine (SVM)

- 1 Shared Pipeline**
Same feature engineering + encoding as Random Forest
- 2 Feature Scaling**
StandardScaler: zero mean, unit variance (required for SVM)
- 3 Training Cap**
Max 15,000 samples — SVM scales $O(n^2)$
- 4 Kernel Selection**
Linear kernel — appropriate for very high dimensional sparse one-hot space.
- 5 Model Training**
SVC(kernel='linear', random_state=42) · predict on full test set

SYSTEM ARCHITECTURE

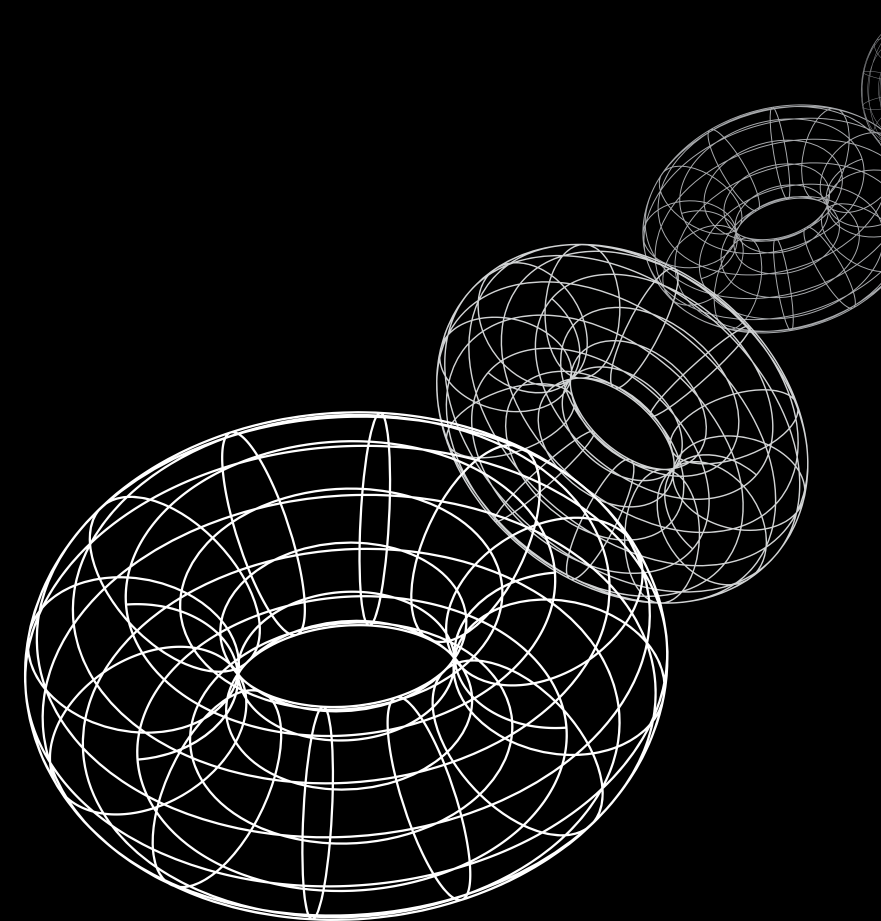
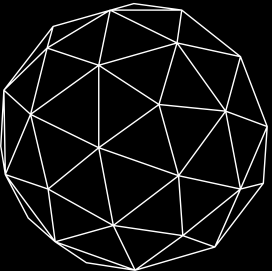


Key Design Decisions: Both models share the same preprocessing pipeline. SVM requires StandardScaler; RF does not. random_state=42 ensures full reproducibility.

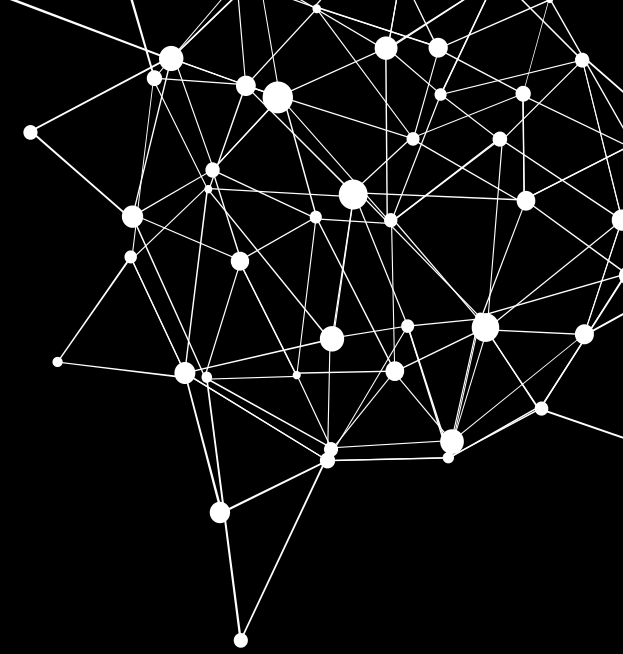


PROJECT CONTRIBUTIONS & INNOVATIONS

- Predicted patient no-shows using machine learning
- Compared Random Forest and SVM performance
- Identified Age and Lead_Time as key predictors
- Improved insights for scheduling and resource planning
- Built a reusable healthcare prediction framework



LIMITATIONS & LESSONS LEARNED



Limitations

- Dataset sourced from a single healthcare organization
- Limited patient information available for analysis
- Historical attendance behaviour was not included
- Models were evaluated offline and not deployed in a live environment

Lessons Learned

- Data quality and preprocessing strongly influence results
- Feature engineering improved predictive performance
- Random Forest outperformed SVM for this dataset
- Multiple evaluation metrics provide a more complete assessment than accuracy alone



RESULTS & EVALUATION

Predicting Patient No-Shows: The Imbalance Problem

- 169,524 appointment records | 33 features
- 80/20 train-test split (random seed = 42)
- 85% Show vs. 15% No-Show

A patient no-show is the minority class and thus the hardest (and most important clinically) outcome to predict. The models naturally tend to predict "Show", as that minimises overall error. This directly harms real-world usefulness.

■ Show — 85% ■ No-show — 15%



169,524
total appointments

33
engineered features

RANDOM FOREST VS SVM : ACCURACY ≠ PERFORMANCE

Comparison table:

Metric	Random forest	SVM
Accuracy	83.72%	85.28%
No-show precision	15.42%	0.00%
No-show recall	2.36%	0.00%
No-show F1-score	0.04	0.00
True positives (no-show)	118	0

Selected Model: Random Forest
Reason: Only model that detected no-shows.

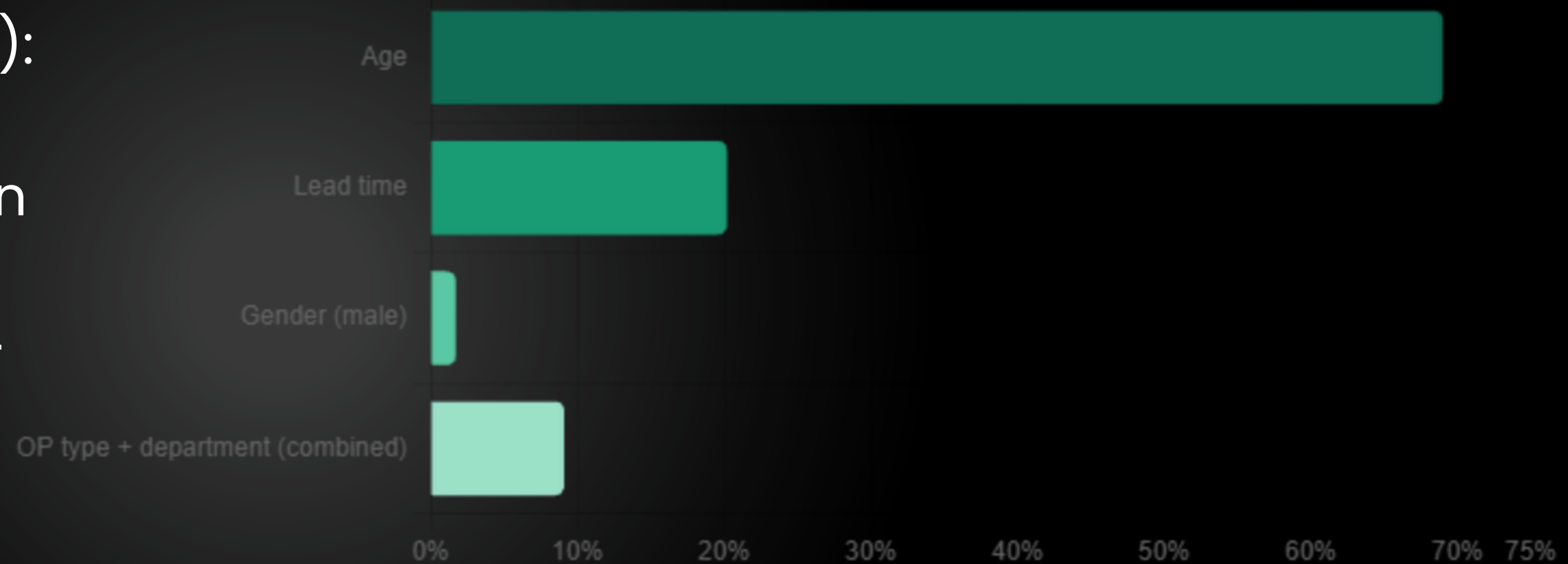
SVM gets higher accuracy, but is a degenerate classifier – it predicted every single patient as “Show”, detecting zero no-shows. Despite lower accuracy, Random Forest is the only model with any real predictive signal for the minority class.

WHAT ACTUALLY DRIVES NO-SHOWS

Key Drivers & Next Steps

Feature importance (Random Forest):

- Age — 69% (dominant predictor)
- Lead Time — 20.2% (days between booking & appointment)
- Gender + Department/OP Type — remaining ~10%



Recommendations:

Apply class-balancing (SMOTE, class weights)

Retrain SVM on the full dataset with `class_weight='balanced'`

Try XGBoost/LightGBM

Lower the classification threshold to boost recall

RECOMMENDATIONS

Implementation

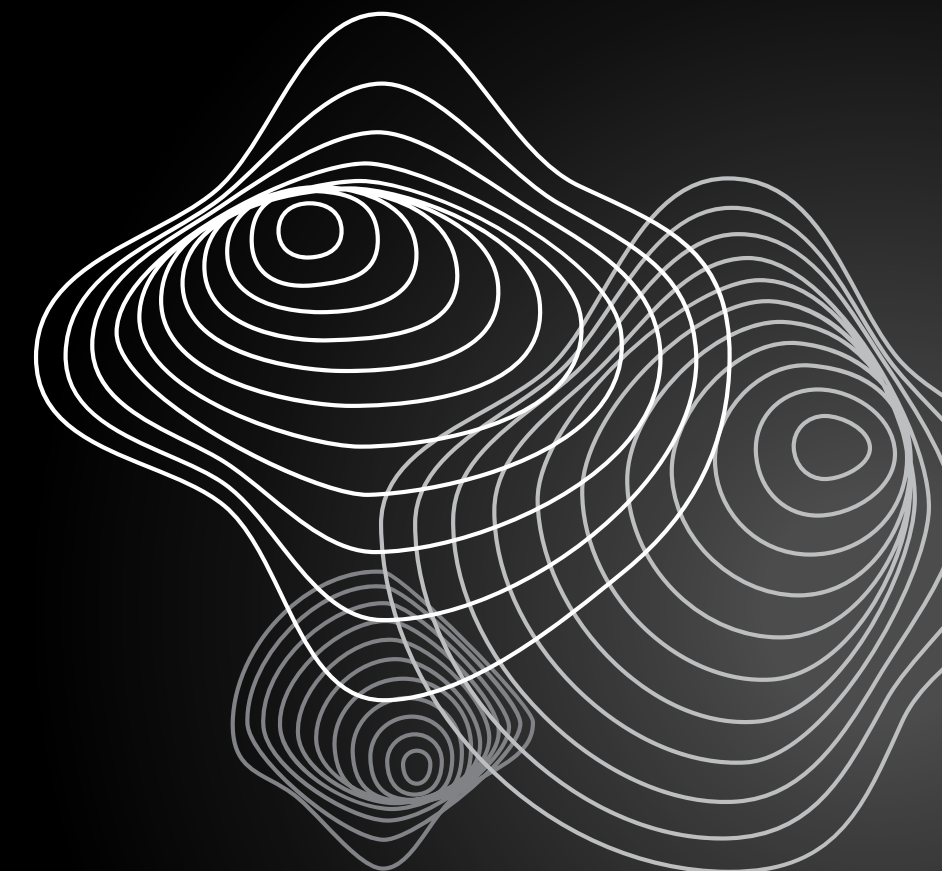
- Integrate the model into the appointment scheduling system
- Identify high-risk patients before appointments
- Send targeted reminders to reduce missed appointments

Data Management

- Improve appointment and attendance data quality
- Track attendance history and reminder responses

Expected Benefits

- Potential reduction in missed appointments
- Better resource utilization
- Improved patient experience
- Increased operational efficiency



FUTURE WORK



Model Enhancements

- Use SMOTE for class balancing
- Evaluate XGBoost and LightGBM models

Scalability

- Multi-hospital deployment
- Adaptation for different departments

Future Directions

- LLM-powered patient communication
- Reinforcement learning for scheduling
- Continuous model monitoring and retraining

CURRENT PREDICTION
FRAMEWORK



ENHANCED FEATURES



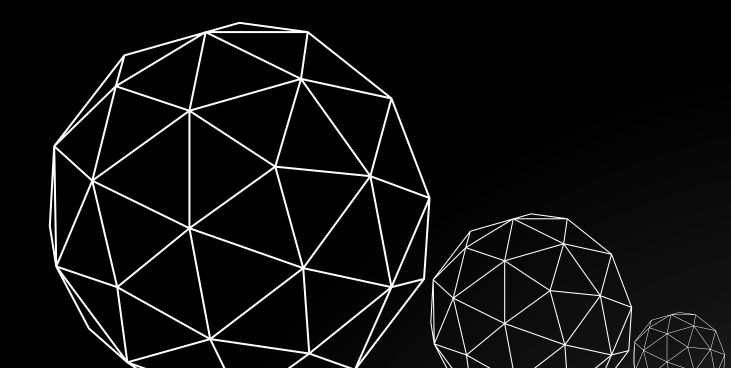
ADVANCED ML MODELS
(XGBOOST / LIGHTGBM)



MULTI-HOSPITAL
VALIDATION



AI-POWERED SCHEDULING
SYSTEM



CONCLUSION

Key Takeaways

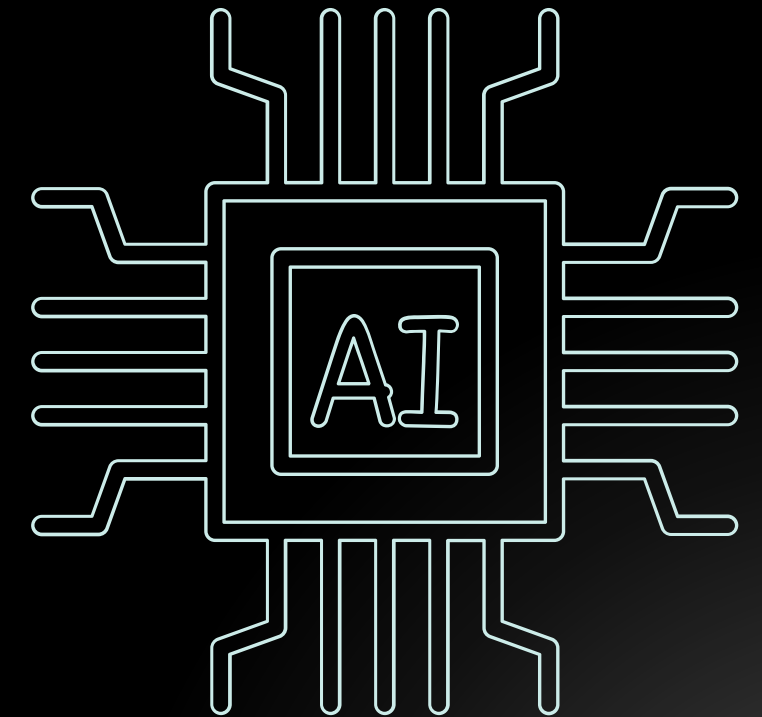
- Machine learning can support appointment attendance analysis
- Random Forest outperformed SVM in model evaluation
- Age and Lead Time showed predictive potential

Project Objectives Achieved

- Developed and evaluated machine learning models
- Identified factors affecting appointment attendance
- Demonstrated AI's potential in healthcare scheduling

Industry Impact & Next Steps

- Demonstrates AI's potential to improve healthcare efficiency and patient management
- Validate the framework using real attendance data
- Continue model monitoring and improvement





Team Members

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Project

AI-Based Hospital No-Show Prediction
System

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THANK YOU